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Q1)

```
#include <iostream>
```

```
#include <vector>
```

```
#include <algorithm>
```

```
using namespace std;
```

```
int BinarySearch(vector<int> &arr, int target)
```

```
{
```

```
    int start = 0;
```

```
    int end = arr.size() - 1;
```

```
    while (start <= end)
```

```
    {
```

```
        int middle = (start + end) / 2;
```

```
        if (arr[middle] == target)
```

```
        {
```

```
            return middle;
```

```
        }
```

```
        if (arr[middle] < target)
```

```
        {
```

```
            start = middle + 1;
```

```
        }
```

```
    else
```

```
    {
```

```
        end = middle - 1;
```

```
    }
```

```
}
```

```
return -1;
```

```
}
```

```
int main()
```

```
{
```

```
    vector<int> arr = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91};
```

```
    int target = 56;
```

```
    int index = BinarySearch(arr, target);
```

```
    cout << index;
```

```
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments> & 'c:\Users\harsh\.vscode\externalConsole.exe' '--stdin=Microsoft-MIEngine-In-ywxh4pv.zgh' '--stdout=Microsoft-MIEngine-Out-ywxh4pv.zgh' --id=Microsoft-MIEngine-Pid-am4b42xf.xc2' '--dbgExe=C:\winlibs-x86_64-posix-seh-gcc-14.2.0\bin\g++.exe' --er=mi'
```

```
5
```

```
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments>
```

Q2)

```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
void merge(vector<int> &arr, int start, int mid, int end)
```

```
{
```

```
    vector<int> temp;
```

```
    int i = start;
```

```
    int j = mid + 1;
```

```
    while (i <= mid && j <= end)
```

```
    {
```

```
        if (arr[i] < arr[j])
```

```
        {
```

```
            temp.push_back(arr[i]);
```

```
            i++;
```

```
        }
```

```
    else
```

```
    {
```

```
        temp.push_back(arr[j]);
```

```
        j++;
```

```
    }
```

```
}
```

```
while (i <= mid)
```

```
{
```

```
    temp.push_back(arr[i]);
```

```
    i++;
```

```
}
```

```
while (j <= mid)
```

```

{
    temp.push_back(arr[j]);
    j++;
}

for (int index = 0; index < temp.size(); index++)
{
    arr[index + start] = temp[index];
}
}

void MergeSort(vector<int> &arr, int start, int end)
{
    if (start < end) /// run till just before start and end index become equal
    {
        int mid = start + (end - start) / 2;
        MergeSort(arr, start, mid); // Left half
        MergeSort(arr, (mid + 1), end); // Right Half

        merge(arr, start, mid, end);
    }
}

int main()
{
    vector<int> arr1 = {12, 11, 13, 5, 6, 7};
    MergeSort(arr1, 0, arr1.size() - 1);

    for (int i = 0; i < arr1.size(); i++)
    {
        cout << arr1[i] << " ";
    }
}

```

```

    }

    cout << endl;

    vector<int> arr2 = {38, 27, 43, 3, 9, 82, 10};

    MergeSort(arr2, 0, arr2.size() - 1);

    for (int i = 0; i < arr2.size(); i++)
    {
        cout << arr2[i] << " ";
    }

    return 0;
}

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments> & 'c:\Users\harsh\.vscode\extensions\ms-vscode
auncher.exe' '--stdin=Microsoft-MIEngine-In-svvpvgyk.ehj' '--stdout=Microsoft-MIEngine-Out-n1iyjzif
id=Microsoft-MIEngine-Pid-dclqjvm1.lrv' '--dbgExe=C:\winlibs-x86_64-posix-seh-gcc-14.2.0-llvm-19.1.
er=mi'
5 6 7 11 12 13
3 9 10 27 38 43 82
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments>

```

Q3)

```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
int partition(vector<int> &arr, int start, int end)
```

```
{
```

```
    int idx = start - 1;
```

```
    int pivot = arr[end];
```

```
    for (int j = start; j < end; j++)
```

```
    {
```

```
        if (arr[j] <= pivot)
```

```
        {
```

```
            idx++;
```

```
            int temp = arr[j];
```

```
            arr[j] = arr[idx];
```

```
            arr[idx] = temp;
```

```
        }
```

```
    }
```

```
    idx++;
```

```
    int temp = arr[end];
```

```
    arr[end] = arr[idx];
```

```
    arr[idx] = temp;
```

```
    return idx;
```

```
}
```

```
void QS(vector<int> &arr, int start, int end)
```

```
{
```

```

if (start < end)
{
    int pivIndex = partition(arr, start, end);

    QS(arr, start, pivIndex - 1);

    QS(arr, pivIndex + 1, end);
}
}

```

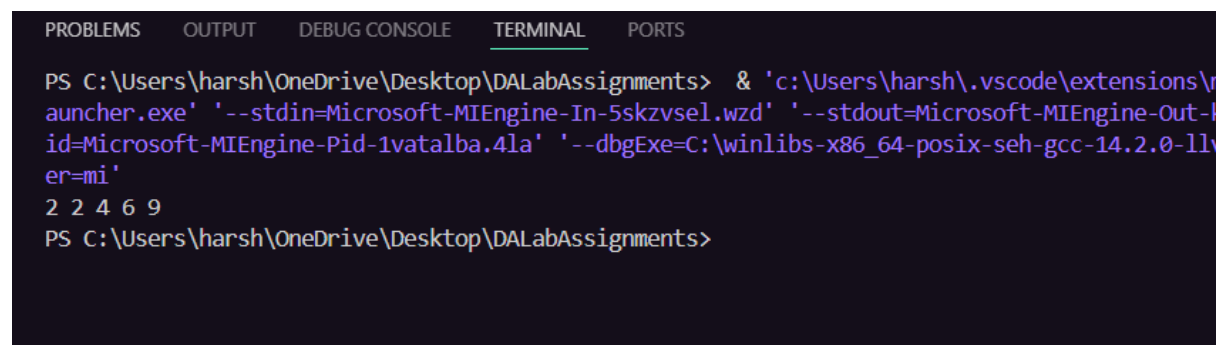
```

int main()
{
    vector<int> arr = {4, 2, 6, 9, 2};

    QS(arr, 0, arr.size() - 1);

    for (int val : arr)
    {
        cout << val << " ";
    }
}

```



```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments> & 'c:\Users\harsh\.vscode\extensions\ms-vscode.cpptools\bin\Debug\cpptools_launcher.exe' '--stdin=Microsoft-MIEngine-In-5skzvsel.wzd' '--stdout=Microsoft-MIEngine-Out-1id=Microsoft-MIEngine-Pid-1vatalba.4la' '--dbgExe=C:\winlibs-x86_64-posix-seh-gcc-14.2.0-11\bin\g++.exe'
2 2 4 6 9
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments>

```

Q4)

```
#include <iostream>
```

```
#include <vector>
```

```
#include <algorithm>
```

```
using namespace std;
```

```
int maxSubarraySum(vector<int> &arr)
```

```
{
```

```
    int max_current = arr[0];
```

```
    int max_global = arr[0];
```

```
    for (int i = 1; i < arr.size(); i++)
```

```
    {
```

```
        max_current = max(arr[i], arr[i] + max_current);
```

```
        max_global = max(max_global, max_current);
```

```
    }
```

```
    return max_global;
```

```
}
```

```
int main()
```

```
{
```

```
    vector<int> arr = {-2, -5, 6, -2, -3, 1, 5, -6};
```

```
    int max = maxSubarraySum(arr);
```

```
    cout << max;
```

```
}
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments> & 'c:\Users\harsh\.vscode\bin\mscp-launcher.exe' '--stdin=Microsoft-MIEngine-In-xmgdn2b2.4xf' '--stdout=Microsoft-MIEngine-Out-xmgdn2b2.4xf' --id=Microsoft-MIEngine-Pid-2zrqkag2.acn' '--dbgExe=C:\winlibs-x86_64-posix-seh\bin\gdb.exe' --er=mi'
```

7

```
PS C:\Users\harsh\OneDrive\Desktop\DALabAssignments>
```